Symmetric polynomials

$$\forall n \in \mathbb{P}$$

$$\forall (R, 0, 1, \oplus, \odot) \ni \forall x \exists \tilde{x} \ni x \oplus \tilde{x} = 0$$

$$R[x_i]_{i=1}^n = \{\text{polynomials with coefficients in } R \text{ and variables } x_1 \cdots x_n\}$$

Monomials

$$n \in \mathbb{P}, \forall i, \alpha_i \in \mathbb{Z}^+, \prod_{i \le n} x_i^{\alpha_i}$$

$$\alpha := \langle a_i \rangle_{i=1}^n$$

$$\alpha \in (\mathbb{Z}^+)^n$$

FACT:

Let,

$$f(x_1, \dots, x_n) \in R[x_i]_{i=1}^n$$

Then, f can be rewritten as

$$\sum_{\alpha \in \mathbb{Z}_+^n} c_\alpha \cdot x^\alpha \ni \forall \alpha, c_\alpha \in R$$

Definition

A polynomial $f(x_1, \dots, x_n) \in R[x_i]_{i=1}^n$ is called symmetric if:

$$\forall X, Y := \langle x_i \rangle_{\substack{i \in X \\ X \in \mathcal{S}([n])}}, fX = fY$$

For example:
Let f be a polynomial of 2 variables. Then f is symmetric if: $f(x, y) = f(y, x)$ - $\forall a, b \in \mathbb{P}, c_1, c_2 \in R f(x, y) = c_1(x^a y^b + y^b x^a) + c_2 xy$

Let f be a polynomial of 3 variables. Then f is symmetric if: $f(x, y, z) = f(x, z, y) = f(y, x, z) = f(y, z, x) = f(z, x, y) = f(z, y, x)$

Proposition

Let $f(X) = \sum_{\alpha \in \mathbb{Z}_+^n} c_\alpha X^\alpha$ be a polynomial then f is symmetric $\iff \forall \lambda, \gamma \in S(\alpha), c_\lambda = c_\gamma$